Sunya Afrasiabi
sunyaafr@engineering.upenn.edu | linkedin.com/in/sunya-afrasiabi

Education

University of Pennsylvania, BSE in Mechanical Engineering, BA in Philosophy — Sept 2024 - May 2028

- **Relevant Coursework:** Thermal-Fluids Engineering, Statics & Strength of Materials, Thermodynamics, Dynamics, Computer-Aided Design (CAD), Computer Programming (Python), Linear Algebra, Differential Equations, Probability

University at Buffalo, Gifted Math Program — Sept 2023 - April 2024

- **Relevant Coursework:** College Calculus III, Introduction to Linear Algebra

Experience & Leadership

University of Pennsylvania – Li Group, Undergraduate Researcher — May 2026 - Present

- Research insect-wing-inspired morphing structures, modeling compliant geometries in **Rhino** and **Grasshopper** to study parametric mechanical behavior.
- Translate computational designs into multi-material prototypes using **powder-based 3D printing** for experimental validation.

NASA L'SPACE Proposal Writing & Evaluation Experience, Radiation Subteam Lead — Sept 2025 - Dec 2025

- Developed innovative solution to the limitation of radiation-resistant materials in HDPE composites to enhance material performance under lunar radiation conditions
- Gained experience in technical proposal composition, project evaluation, and NASA's mission review process.
- Ranked in the **Top 6 out of 40+ national teams**.

Penn Astronomical Student Association, Outreach & Marketing Director — Sept 2025 - Present

- Lead outreach initiatives and external partnerships, securing **$200** in sponsorship funding.
- Configure and calibrate equatorial telescopes and optical tracking systems for public observation events.

#MyStory Program, Founder and Project Manager — Jan 2021 - Aug 2024

- Founded and managed a non-profit oral history initiative promoting cultural awareness through exhibitions.
- Secured over **$24,000** via targeted donations, fundraisers, and corporate grant programs (including Riley's Way Foundation, Prudential, Ashoka, and the Peter & Elizabeth Tower Foundation).
- Directed operations for **20+ community partners** and **40+ volunteers**.

Technical Projects

Multi-Phase Butane Rocket & Numerical Analysis — Spring 2026

- Designed and optimized a variable-mass butane-powered water rocket using aerodynamic modeling, thermodynamic analysis, and experimental validation to navigate constrained flight paths and minimize target error.
- Developed a custom Python Euler integration solver coupling butane vaporization, pressure-driven propulsion, drag, and ballistic dynamics; implemented trajectory reconstruction and multi-variable optimization to improve predictive accuracy.

Mars Rover Chassis Prototype — Spring 2026

- Designed and optimized a Mars rover chassis prototype for University Rover Challenge (URC) constraints as part of a 3-person engineering team, developing detailed SolidWorks assemblies and structural architectures.
- Performed FEA-driven design iterations to optimize load paths, bracket placement, wall thicknesses, and lightweight geometries while balancing structural integrity and mass reduction.

Vacuum-Powered Agricultural Seed Cleaner — Fall 2025

- Designed and fabricated a vertical pneumatic seed-chaff separator with a zig-zag separation column, custom swirl chamber, laser-cut acrylic panels, and 3D-printed nozzles to optimize airflow behavior and separation efficiency.

Papers

Linear Algebra Foundations of Neural Networks, University of Pennsylvania — August 2025

- Wrote expository research paper connecting linear algebra, neural networks, and robotics. **(Link)**

Aviation Solutions for Wildlife Prevention, Wharton Undergraduate Aerospace Club — May 2025

- Co-authored investor-oriented whitepaper on UAVs, AI wildfire prediction, and aerospace innovations, framing $6.4B+ market opportunities and potential $5B annual cost savings. **(Link to File)**

Skills

Software: Rhino 3D, Grasshopper, SolidWorks, MATLAB, Python, NumPy, FEA
Fabrication: Laser Cutting, FDM/SLA 3D Printing, Mechanical Assembly
Additional: Computational Geometry, Bio-Inspired Design, Technical Writing, Logic, Project Management